

Experiment. no: 4
flip flops

Aim: - To familiarize with the working of
JK Flipflop and D. Flipflop using ICs
7473 and 7474 respectively.

Components Required: - IC 7473, IC 7474,
patch cords, IC trainer kit.

THEORY

Logic circuits that incorporate memory cells are called sequential logic circuits. Their output depends ~~depend~~ not only upon the present value of the input but also upon the previous values. Sequential logic circuits often require a timing generator (clock) for their operation. The latch (flipflop) is a basic bistable memory element widely used in sequential logic circuits. Usually there are two outputs Q and complementary value $\bar{Q}$.

IC 7473 is a dual, negative-edge triggered JK Flipflop with individual clear inputs. It contains two independent JK Flip-flops, each capable of storing a single bit of binary information and featuring complementary outputs. The Flip-flops are considered master-slave and operate on the falling edge of the clock pulse.

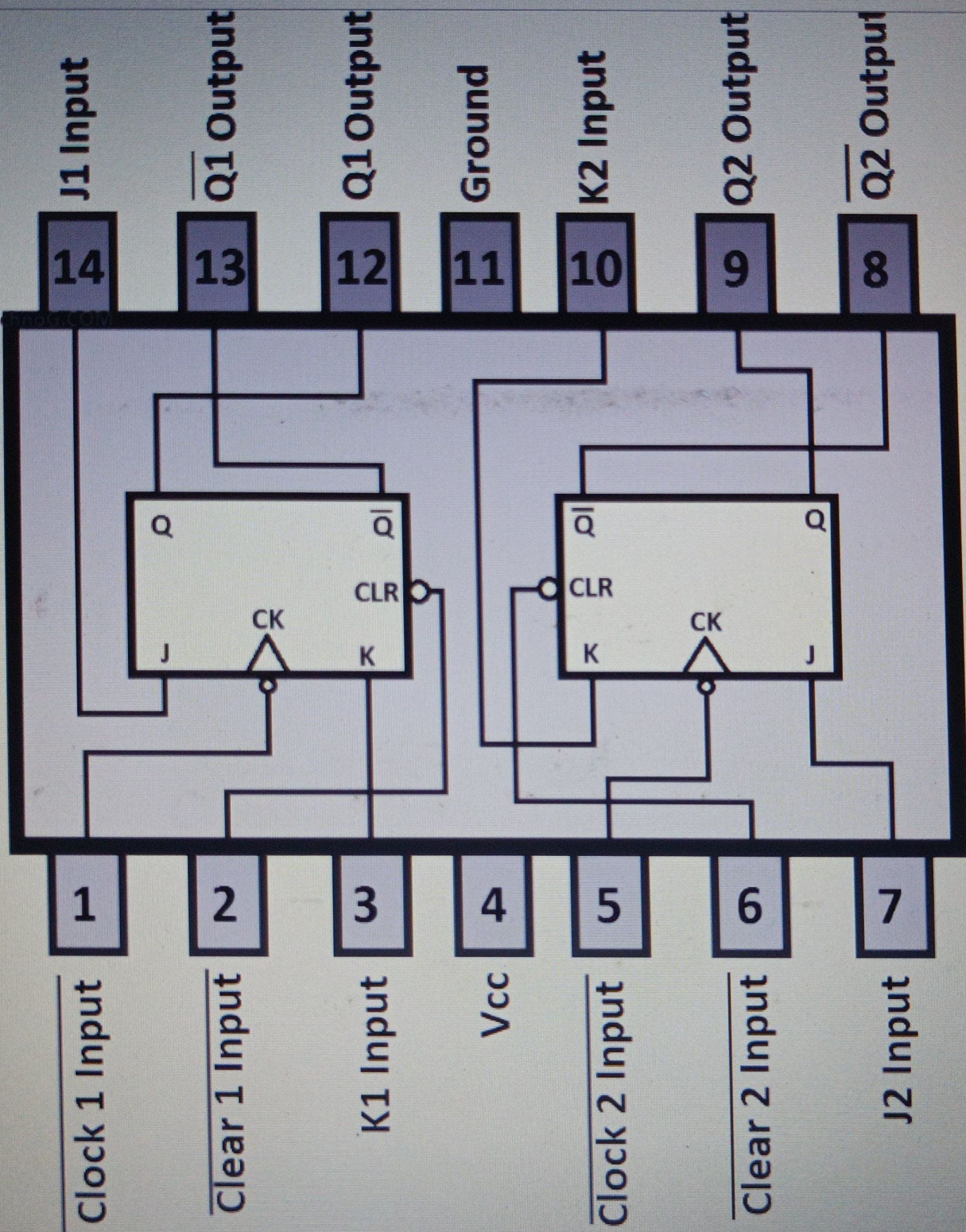
IC 7474 is a versatile dual D-type Flip-Flop IC designed with two separate Flip-Flops that respond to positive clock edges. Each Flip-Flop comes with clear, preset and complementary output features making it useful for storing and managing data in binary form. This IC not only stores data but also allows for changes to the stored data when required, giving it flexibility in digital applications.

PROCEDURE

- Check all the components for their working
- Insert the appropriate IC into the IC base
- Make connections as shown in the circuit diagram
- Verify truth table and observe outputs.

Result

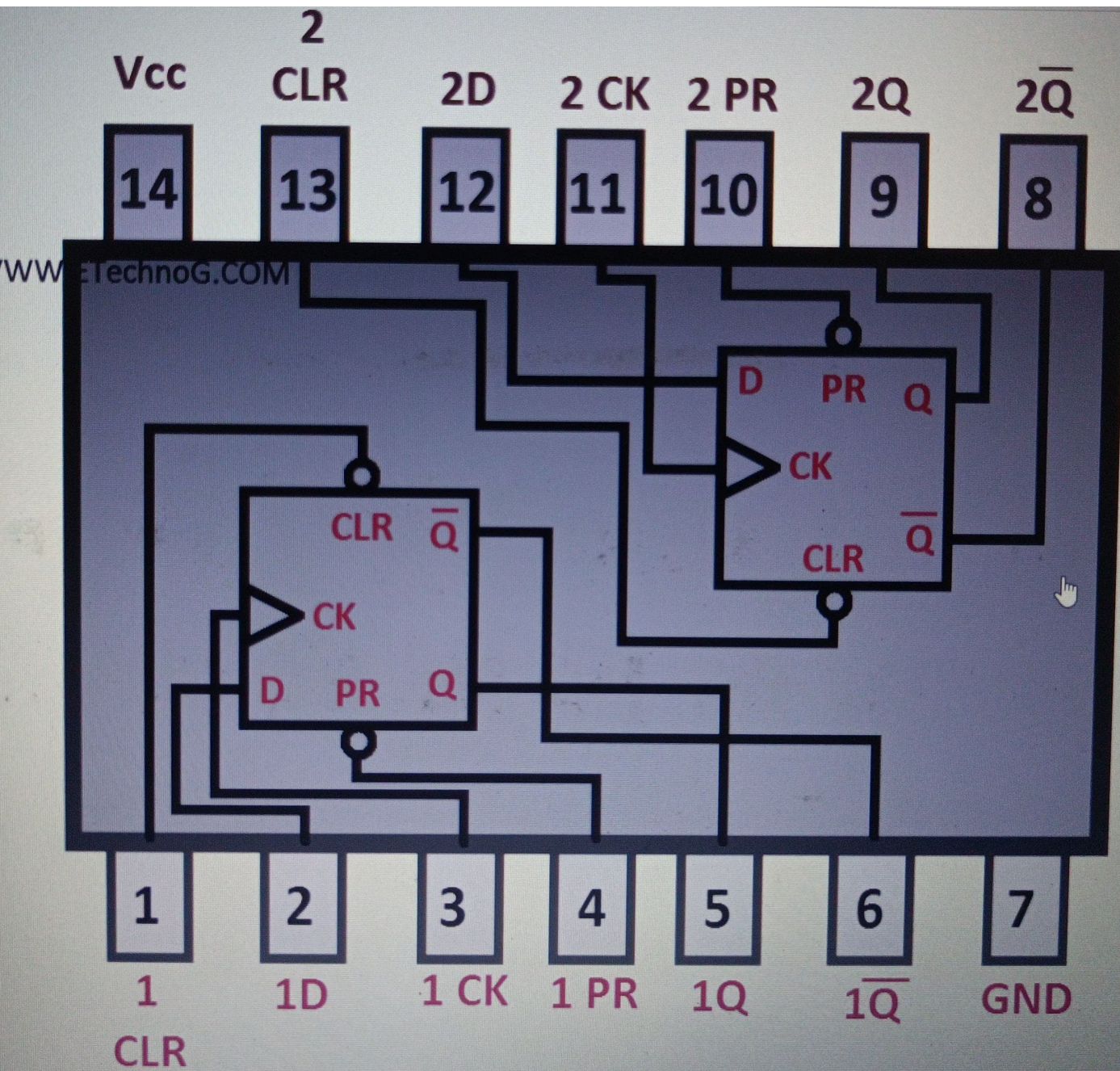
Familiarized with the working
of IC 7473 and IC 7474.



IC 7473 Pinout Diagram

FUNCTION TABLE

INPUTS				OUTPUTS	
$\overline{\text{CLR}}$	CLK	J	K	Q	$\overline{Q}$
L	X	X	X	L	H
H	↓	L	L	Q_0	$\overline{Q_0}$
H	↓	H	L	H	L
H	↓	L	H	L	H
H	↓	H	H	TOGGLE	
H	H	X	X	Q_0	$\overline{Q_0}$



IC 7474 Pinout Diagram

FUNCTION TABLE

INPUTS				OUTPUTS	
$\overline{\text{PRE}}$	$\overline{\text{CLR}}$	CLK	D	Q	$\overline{\text{Q}}$
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H ⁽¹⁾	H ⁽¹⁾
H	H	↑	H	H	L
H	H	↑	L	L	H
H	H	L	X	Q ₀	$\overline{\text{Q}}_0$

- (1) This configuration is nonstable; that is, it does not persist when $\overline{\text{PRE}}$ or $\overline{\text{CLR}}$ returns to its inactive (high) level.